WORKED SOLUTION

STEP III 1997, Question 12

Probability and Statistics • Coupon Collector Problem

STEP III 1997, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q12

The key idea

Part (i) is the geometric distribution, and its probability generating function hands us the mean and variance by differentiation: $\mathrm{E}(K) = \mathrm{G}'(1)$ and $\mathrm{E}(K(K-1)) = \mathrm{G}''(1)$. In part (ii), the trick is to split the long wait for a full set into 52 short waits, one for each *new* card. Each short wait is a "toss until the first head" experiment, so part (i) does all the work.

Solution

Part (i)

The first head comes on toss k exactly when the first $k-1$ tosses are tails and toss k is a head. The tosses are independent, so

$$\mathrm{P}(K = k) = q^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

Hence

$$\mathrm{G}(s) = \sum_{k=1}^{\infty} pq^{k-1}s^k = ps\sum_{j=0}^{\infty}(qs)^j = \frac{ps}{1-qs},$$

summing the geometric series, which converges for $|qs| < 1$ (in particular for s near 1 when $p > 0$). That is the required result.

Differentiating,

$$\mathrm{G}'(s) = \frac{p(1-qs) + psq}{(1-qs)^2} = \frac{p}{(1-qs)^2},$$
$$\mathrm{G}''(s) = \frac{2pq}{(1-qs)^3}.$$

Since $\mathrm{G}'(s) = \sum k\,\mathrm{P}(K=k)s^{k-1}$ and $\mathrm{G}''(s) = \sum k(k-1)\mathrm{P}(K=k)s^{k-2}$, putting $s = 1$ and using $1 - q = p$ gives

$$\mathrm{E}(K) = \mathrm{G}'(1) = \frac{p}{p^2} = \frac{1}{p}, \qquad \mathrm{E}(K(K-1)) = \mathrm{G}''(1) = \frac{2pq}{p^3} = \frac{2q}{p^2}.$$

Therefore

$$\begin{aligned}\mathrm{Var}(K) &= \mathrm{E}(K(K-1)) + \mathrm{E}(K) - (\mathrm{E}(K))^2 \\ &= \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{2q+p-1}{p^2} = \frac{q}{p^2}.\end{aligned}$$

So $\mathrm{E}(K) = \dfrac{1}{p}$ and $\mathrm{Var}(K) = \dfrac{q}{p^2}$.

Part (ii)

Let N_i be the number of draws made after we have seen exactly $i-1$ different cards, up to and including the draw that shows the ith different card. So $N_1 = 1$ (the first card is always new), and

$$N = N_1 + N_2 + \cdots + N_{52}.$$

Ask yourself Why is each N_i a "first head" variable like K?

Once we have seen $i-1$ different cards, $53-i$ of the 52 cards are still unseen. Because we draw with replacement, every draw is independent of the past and shows an unseen card with the same probability

$$p_i = \frac{53-i}{52}.$$

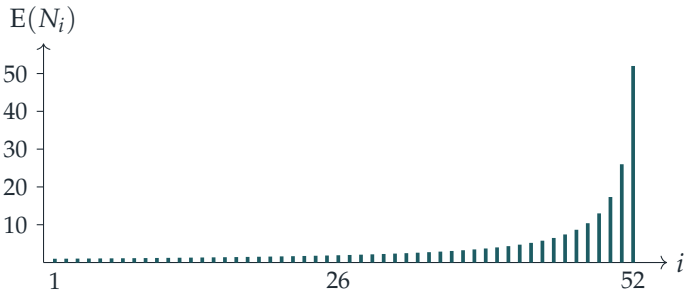
Call such a draw a "head". Then N_i is the number of draws up to and including the first head, which is exactly the variable K of part (i) with $p = p_i$. Hence

$$\mathrm{E}(N_i) = \frac{1}{p_i} = \frac{52}{53-i}.$$

Expectation is linear (we don't need independence of the N_i for this), so

$$\begin{aligned}\mathrm{E}(N) &= \sum_{i=1}^{52} \frac{52}{53-i} = 52\left(\frac{1}{52} + \frac{1}{51} + \cdots + \frac{1}{2} + \frac{1}{1}\right) \\ &= 52\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{52}\right).\end{aligned}$$

The early waits are short and the last few dominate: the very last card alone takes 52 draws on average.



To estimate, use $1+\frac{1}{2}+\cdots+\frac{1}{n}\approx 0.5+\ln n$ with $n=52$. For $\ln 52$: $2.7^2=7.29$ and $7.29^2\approx 53.1$, so $e^4\approx 53\approx 52$, giving $\ln 52\approx 4$. Therefore

$$\mathrm{E}(N)\approx 52\,(0.5+4)=52\times 4.5=234.$$

So on average we need roughly 234 draws, about four and a half times the size of the pack, to see every card.

Checks

- In (i), $\mathrm{G}(1)=p/(1-q)=1$, as any probability generating function must. With a fair coin ($p=\frac{1}{2}$) we get $\mathrm{E}(K)=2$ and $\mathrm{Var}(K)=2$; and as $p\to 1$ the variance $q/p^2\to 0$, as it should when the first toss is almost certainly a head.
- The exact value is $52H_{52}\approx 235.98$, and a simulation of 10^5 full collections gave a mean of 235.7. Our estimate of 234 is close; the small shortfall comes from the approximation $0.5+\ln n$ (a sharper one is $\ln n+0.577$) and from rounding $\ln 52\approx 3.95$ up to 4.

Common slips

- Getting the success probability of N_i upside down: the chance of a *new* card is $(53-i)/52$, which shrinks as the collection grows.
- Taking $\mathrm{Var}(K)=\mathrm{G}''(1)-\mathrm{G}'(1)^2$, forgetting the $+\mathrm{G}'(1)$ term; $\mathrm{G}''(1)$ is $\mathrm{E}(K(K-1))$, not $\mathrm{E}(K^2)$.
- Thinking the N_i must be independent before adding expectations. Linearity of expectation holds regardless (though here they are in fact independent).

Beyond STEP

This is the *coupon collector's problem*: with n equally likely types, the expected time to collect all of them is $nH_n\approx n\ln n$. Undergraduate courses go further and show that the time is sharply concentrated around $n\ln n$, with fluctuations of order n, and probability generating functions become a standard tool for sums of independent variables like these. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions and the geometric distribution carefully; Blitzstein and Hwang, *Introduction to Probability*, discusses the coupon collector problem with many variations.

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